The leading constant for $S_n \times A$

Let $n \in \{3, 4, 5\}$, and let A be a finite abelian group of order $m = |A|$. Then we consider $G = S_n \times A$. Write $N_G(X)$ for the number of $\mathbb{Q}$-isomorphism classes of degree mn fields L with $\mathrm{Gal}(L^c/\mathbb{Q}) \simeq G$ and $|\operatorname{Disc} L| < X$. All signatures are included. The constant in $N_G(X) \sim C_G X^{1/m}$ as $X \to \infty$ is given by

$$C_G := \frac{\kappa_n}{|\operatorname{Aut} A|} \sum_{B \le A} \mu(B, A) \prod_{p \text{ prime}} \left[(1 - p^{-1}) \sum_{\substack{\psi \in \mathcal{H}_p(B) \\ E \in \mathcal{E}_n(p)}} \frac{p^{-d_p(E \otimes_{\mathbb{Q}_p} L_\psi)/h_\psi}}{|\operatorname{Aut}_{\mathbb{Q}_p}(E)|} \right].$$

Notation. The constants are $\kappa_3 = 1/3$, $\kappa_4 = 5/24$ and $\kappa_5 = 13/120$. The sum is over all subgroups B of A. The Möbius function is defined by

$$\mu(B, A) := \begin{cases} \prod_p (-1)^{r_p} p^{\frac{r_p(r_p-1)}{2}} & \text{if } A/B \cong \prod_p (\mathbb{Z}/p\mathbb{Z})^{r_p}, \\ 0, & \text{otherwise.} \end{cases}$$

For a prime p, the finite set $\mathcal{H}_p(B)$ consists of all continuous group homomorphisms $\psi : \mathbb{Z}_p^\times \to B$. Extend ψ to $\mathbb{Q}_p^\times$ by setting $\psi(p) = 1$, where 1 is the identity of B. Local class field theory assigns to the kernel of this extension a finite abelian extension $L_\psi/\mathbb{Q}_p$ of degree $h_\psi = |\operatorname{im} \psi|$.

Let $\mathcal{E}_n(p)$ be the finite set of isomorphism classes of degree n étale $\mathbb{Q}_p$-algebras. For $E \in \mathcal{E}_n(p)$, let $d_p(E \otimes_{\mathbb{Q}_p} L_\psi)$ be the discriminant exponent of the maximal $\mathbb{Z}_p$-order in $E \otimes_{\mathbb{Q}_p} L_\psi$. Finally, $\operatorname{Aut}_{\mathbb{Q}_p}(E)$ is the group of $\mathbb{Q}_p$-algebra automorphisms of E.

Now, we give some concrete examples and numerics:

Some numerics

n	$C_{S_n \times C_2}$	$C_{S_n \times (C_2 \times C_2)}$
3	0.600485169929...	0.685009888568...
4	0.274102386235...	0.131474579489...
5	0.067505846256...	0.012294591845...

The asymptotic powers are $X^{1/2}$ and $X^{1/4}$ respectively.

The cubic cases

$$C_{S_3 \times C_2} = \frac{1}{3} \left[\frac{68 + 7\sqrt{2}}{64} \prod_{p>2} \left(1 + \frac{2}{p^{3/2}} - \frac{1}{p^{5/2}} - \frac{1}{p^3} - \frac{1}{p^{7/2}} \right) - \frac{7}{8} \prod_{p>2} \left(1 - \frac{1}{p^3} \right) \right].$$

$$\begin{aligned} C_{S_3 \times (C_2 \times C_2)} = \frac{1}{18} \Bigg[& \frac{208 + 45\sqrt{2}}{128} \prod_{p>2} \left(1 + \frac{6}{p^{3/2}} - \frac{3}{p^{5/2}} - \frac{1}{p^3} - \frac{3}{p^{7/2}} \right) \\ & - \frac{3(68 + 7\sqrt{2})}{64} \prod_{p>2} \left(1 + \frac{2}{p^{3/2}} - \frac{1}{p^{5/2}} - \frac{1}{p^3} - \frac{1}{p^{7/2}} \right) \\ & + \frac{7}{4} \prod_{p>2} \left(1 - \frac{1}{p^3} \right) \Bigg]. \end{aligned}$$

The quartic cases

$$C_{S_4\times C_2} = \frac{5}{24}\left[\frac{353}{256}\prod_{p>2}\left(1+\frac{4}{p^2}-\frac{2}{p^3}-\frac{3}{p^4}\right) - \frac{17}{16}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^3}-\frac{1}{p^4}\right)\right].$$

$$\begin{aligned} C_{S_4\times(C_2\times C_2)} = \frac{5}{144}\Bigg[&\frac{560+3\sqrt{2}}{256}\prod_{p>2}\left(1+\frac{10}{p^2}-\frac{4}{p^3}-\frac{7}{p^4}\right) \\ &- \frac{1059}{256}\prod_{p>2}\left(1+\frac{4}{p^2}-\frac{2}{p^3}-\frac{3}{p^4}\right) \\ &+ \frac{17}{8}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^3}-\frac{1}{p^4}\right)\Bigg]. \end{aligned}$$

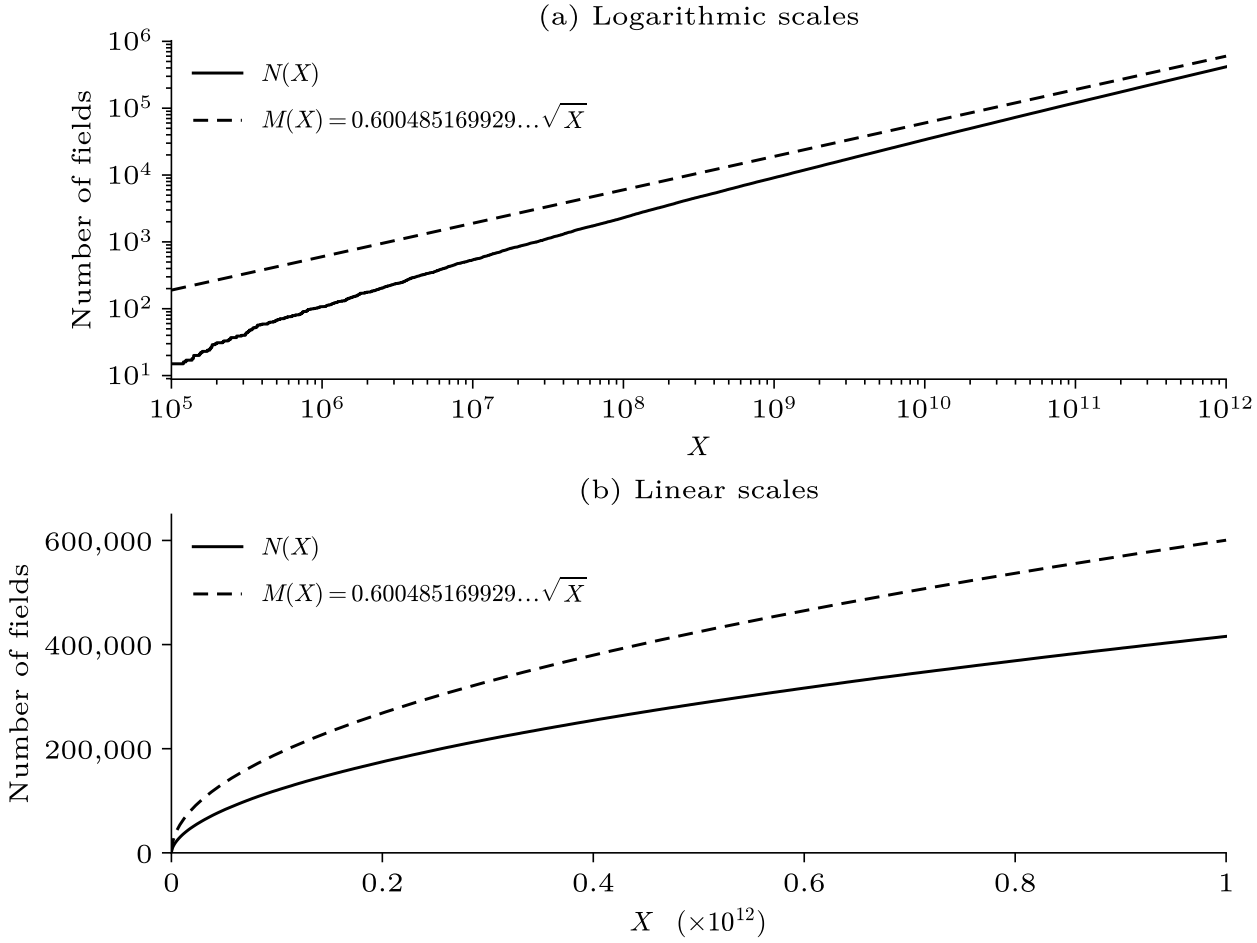
The quintic cases

$$C_{S_5\times C_2} = \frac{13}{120}\left[\frac{648+19\sqrt{2}}{512}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{3}{p^{5/2}}-\frac{1}{p^4}-\frac{2}{p^{9/2}}-\frac{1}{p^5}-\frac{1}{p^{11/2}}\right) - \frac{37}{32}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^4}-\frac{1}{p^5}\right)\right].$$

$$\begin{aligned} C_{S_5\times(C_2\times C_2)} = \frac{13}{720}\Bigg[&\frac{3133+243\sqrt{2}}{2048}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{9}{p^{5/2}}-\frac{1}{p^4}-\frac{6}{p^{9/2}}-\frac{1}{p^5}-\frac{3}{p^{11/2}}\right) \\ &- \frac{3(648+19\sqrt{2})}{512}\prod_{p>2}\left(1+\frac{1}{p^2}+\frac{3}{p^{5/2}}-\frac{1}{p^4}-\frac{2}{p^{9/2}}-\frac{1}{p^5}-\frac{1}{p^{11/2}}\right) \\ &+ \frac{37}{16}\prod_{p>2}\left(1+\frac{1}{p^2}-\frac{1}{p^4}-\frac{1}{p^5}\right)\Bigg]. \end{aligned}$$

The leading constant for $S_n \times A$

For $G = S_3 \times C_2$, the two functions below are the actual count $N(X) = N_{S_3\times C_2}(X)$ and its leading term $M(X) = C_{S_3\times C_2}\sqrt{X} = (0.600485169929\ldots)\sqrt{X}$. The solid curve represents $N(X)$ and the dashed curve represents $M(X)$.



Some numerics

X	$N(X)$	$M(X)$
10^6	108	600.48...
10^8	2299	6004.85...
10^{10}	33770	60048.51...
10^{11}	120217	189890.08...
10^{12}	415740	600485.16...

The counts include all signatures and are complete for $X \le 10^{12}$ by using the PARI/GP cubic tables and all relevant quadratic fields. Their discriminant and signature multiplicities below 10^7 agree with the LMFDB.